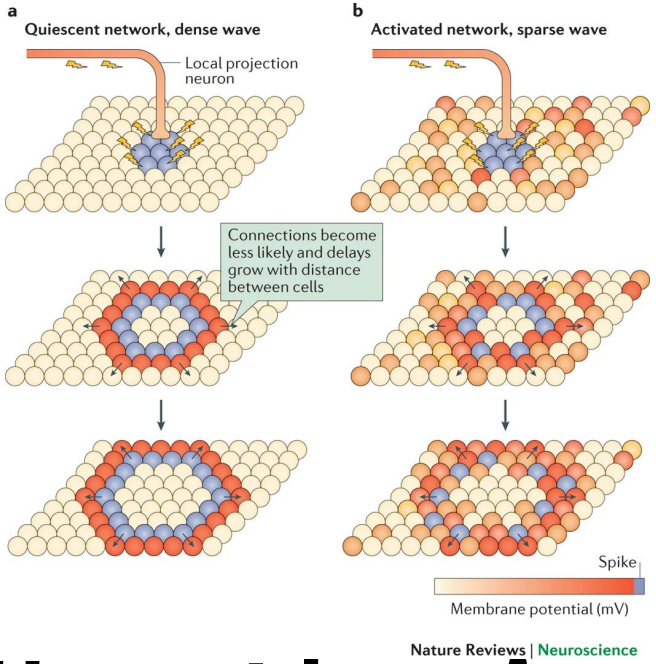
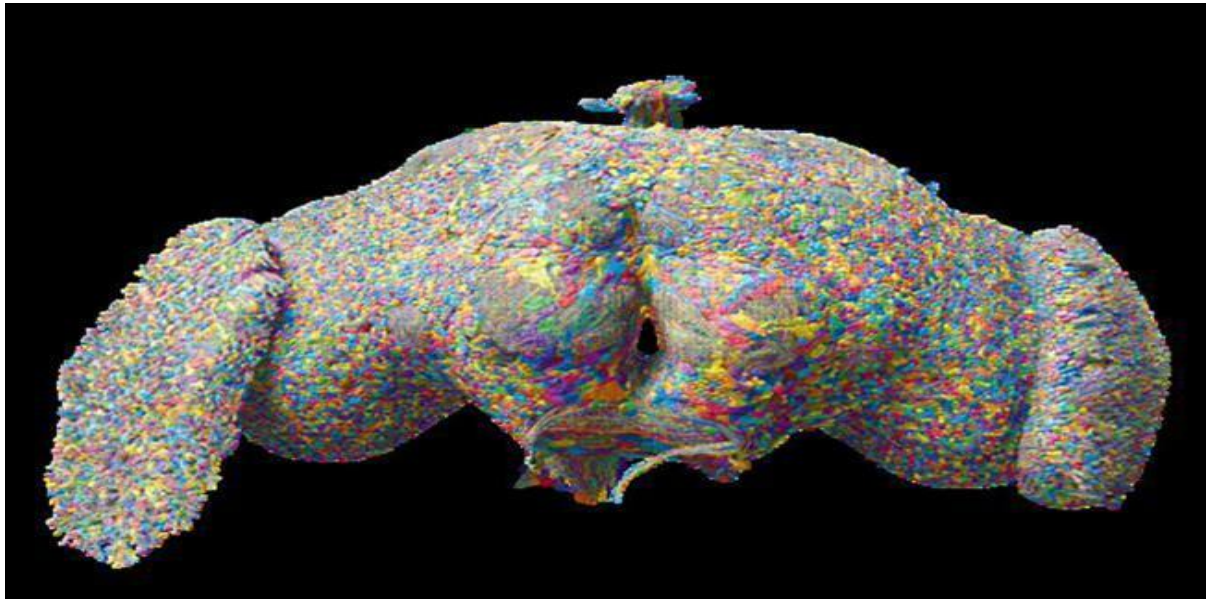
Gay Flies: Connectome-Constrained Visual Simulation with Morphology-Resolved Surface Dynamics



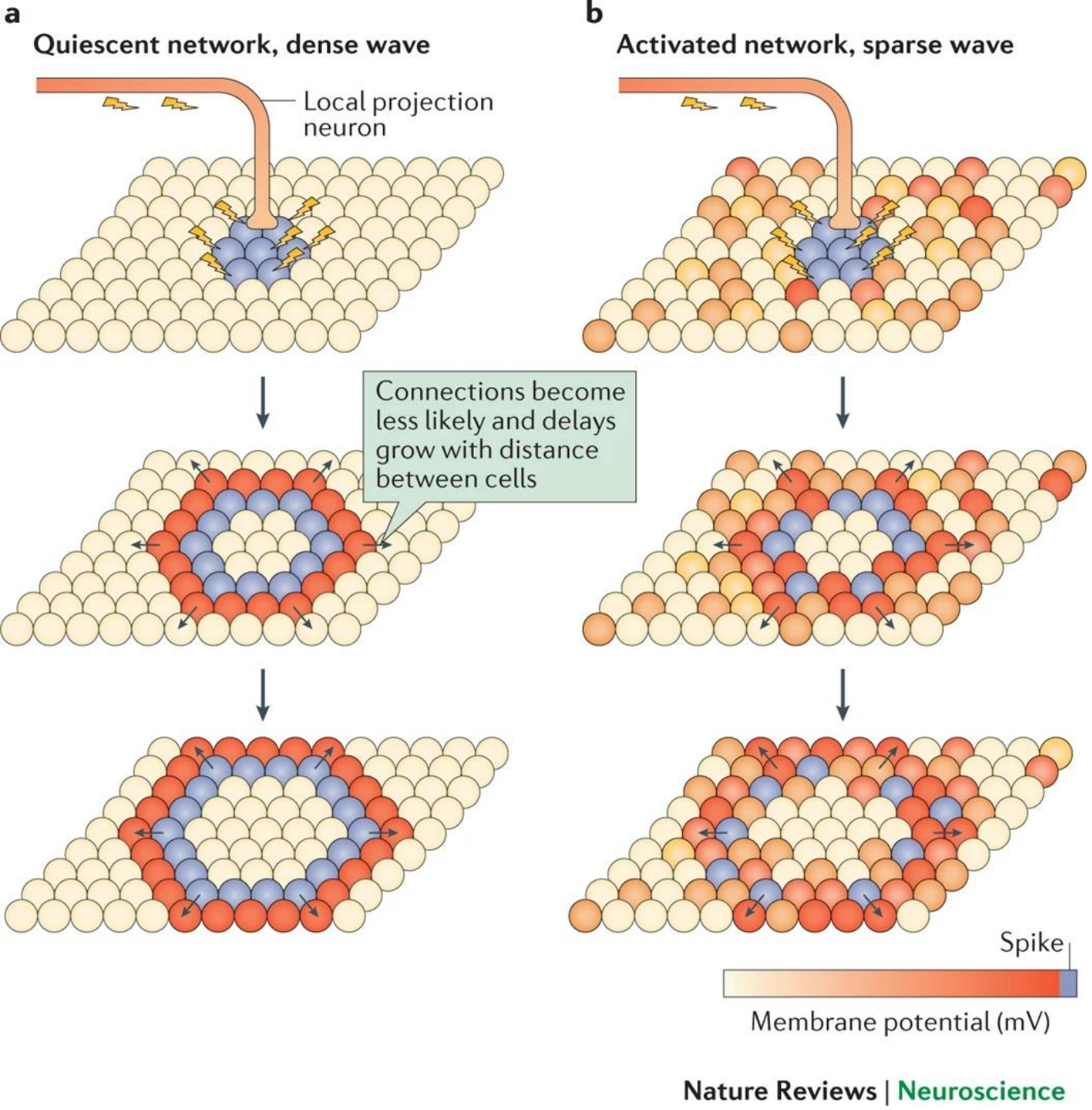
Hypothesis:

Morphology-resolved surface dynamics may constitute an additional intraneuronal computational degree of freedom, allowing the distributed state on neuronal structure to shape information processing.



Fruit(y) Fly Brain:

If we model neurons as having internal surface activity that can move around on their shape, instead of treating them like simple dots, does the circuit do anything measurably different?



Brain Waves Travel…

Science shows that neurons fire in waves across the brain. “Brain Waves”

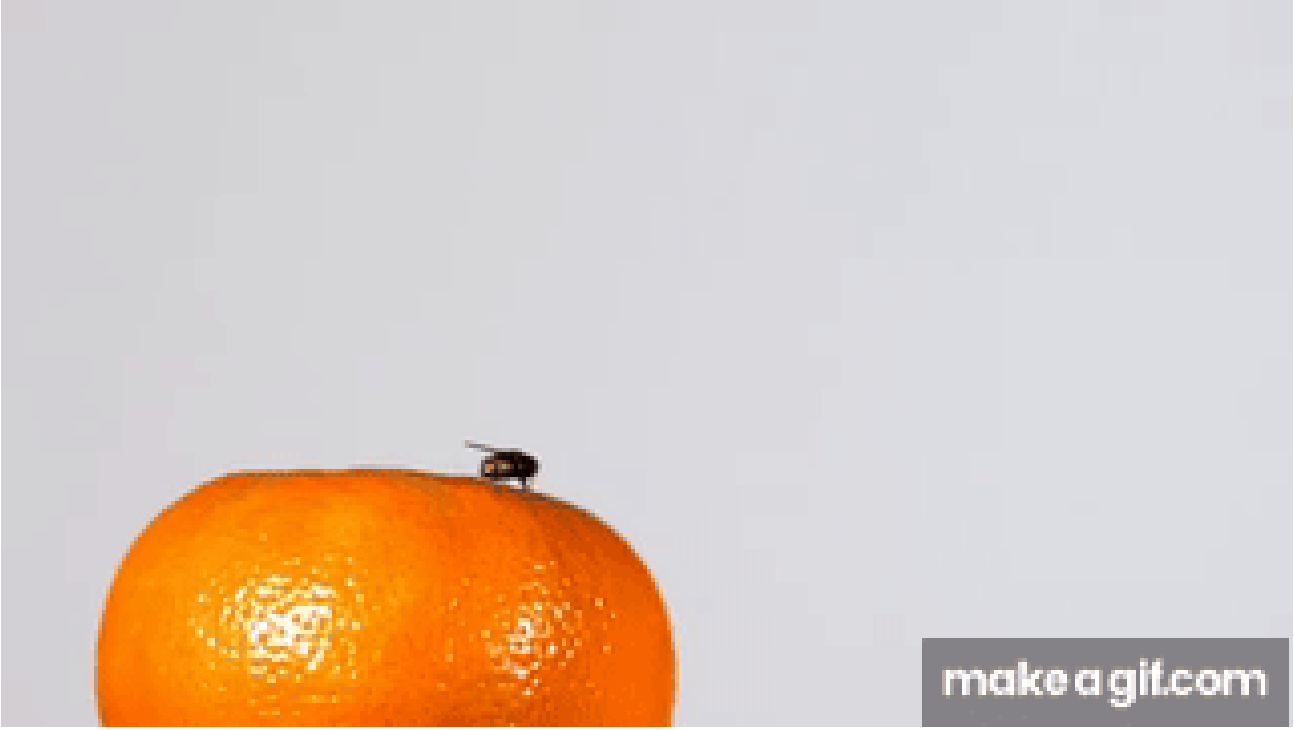
Does Surface Voltage Propagation Affect That?

We’re testing whether similar activity across a single neuron’s shape / size also matters.

Muller, L., Chavane, F., Reynolds, J. *et al.* Cortical travelling waves: mechanisms and computational principles. *Nat Rev Neurosci* **19**, 255–268 (2018). https://doi.org/10.1038/nrn.2018.20

EXPECTED EFFECTS

- Wrong-way motion gets suppressed differently
  - Entropy (will explain, don't worry)
- Response timing shifts slightly
  - Neuron reaction time to new motion (takes time to spread)
- Scrambling the shape modifies the effect
  - This suggests the effect really comes from neuron shape



Comparing:

Point-like Neurons (what's been done)

- Whole brain as **point-like neurons**
- Looks at overall path given a specific input

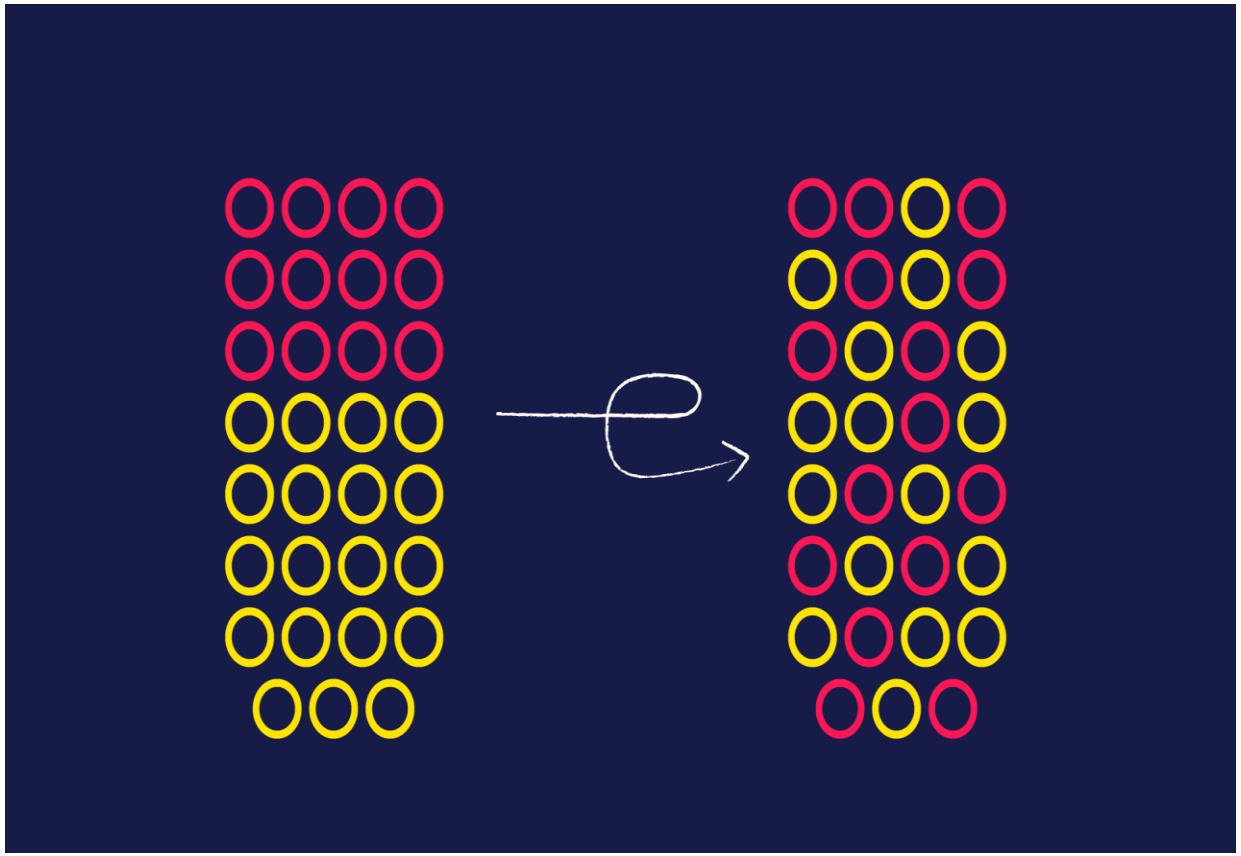
Mesh Surface Adaptation (what were doing)

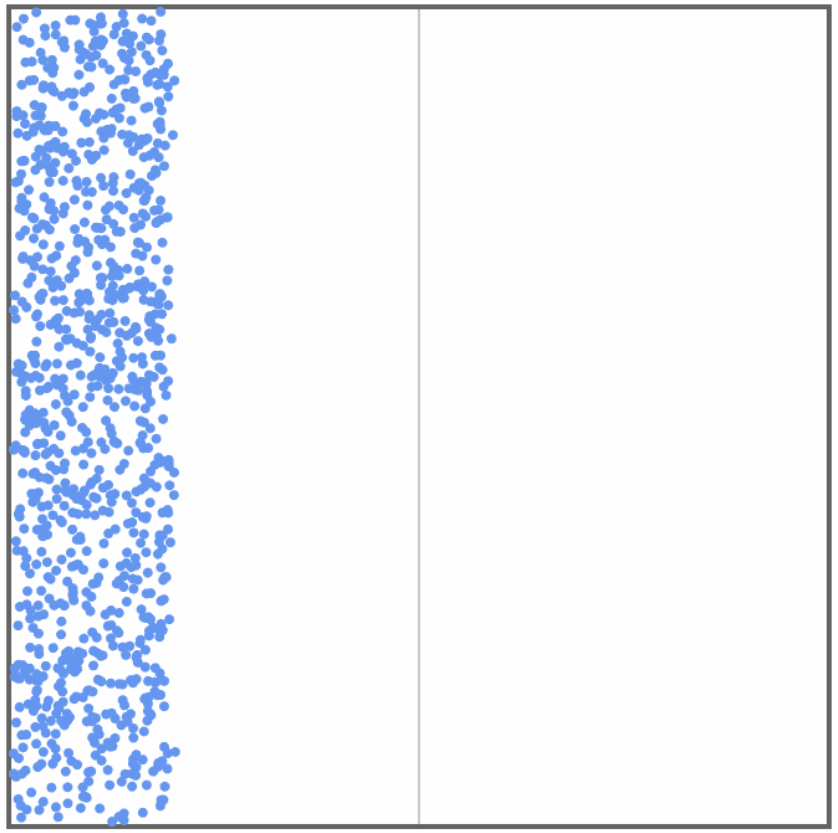
- Whole neuron as **point-like surface areas**
- Looks at the same path but compares the time sensitive effects

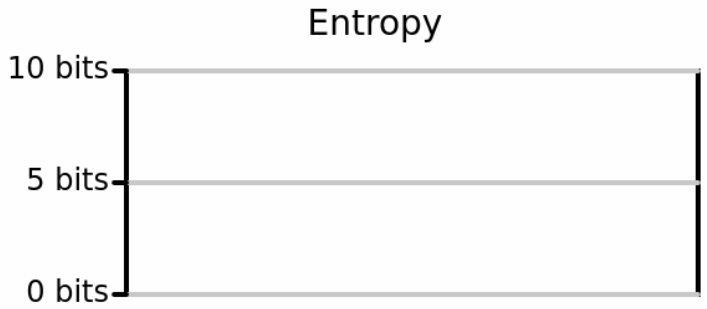
HOW?

Veo

ENTROPY







- THERMODYNAMICS SAYS ENTROPY LIKES TO INCREASE

A.K.A.

- ENERGY LIKES TO SPREAD OUT

PHYSICS

- Waves take time to propagate

We will show you

EXPECTED EFFECTS

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LASTLY

- Big shape
- Big propagation
- Small Shift in time
- Because neurons are small

REMEMBER?

EXPECTED EFFECTS

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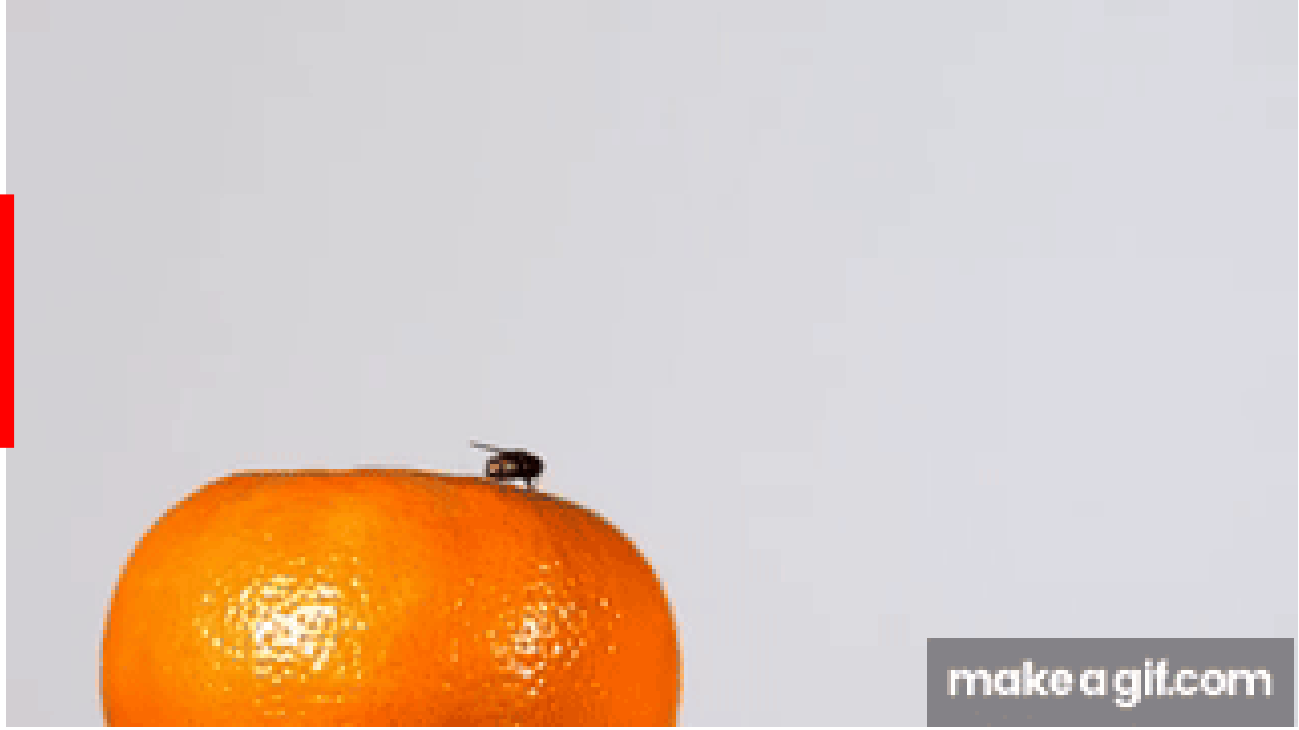
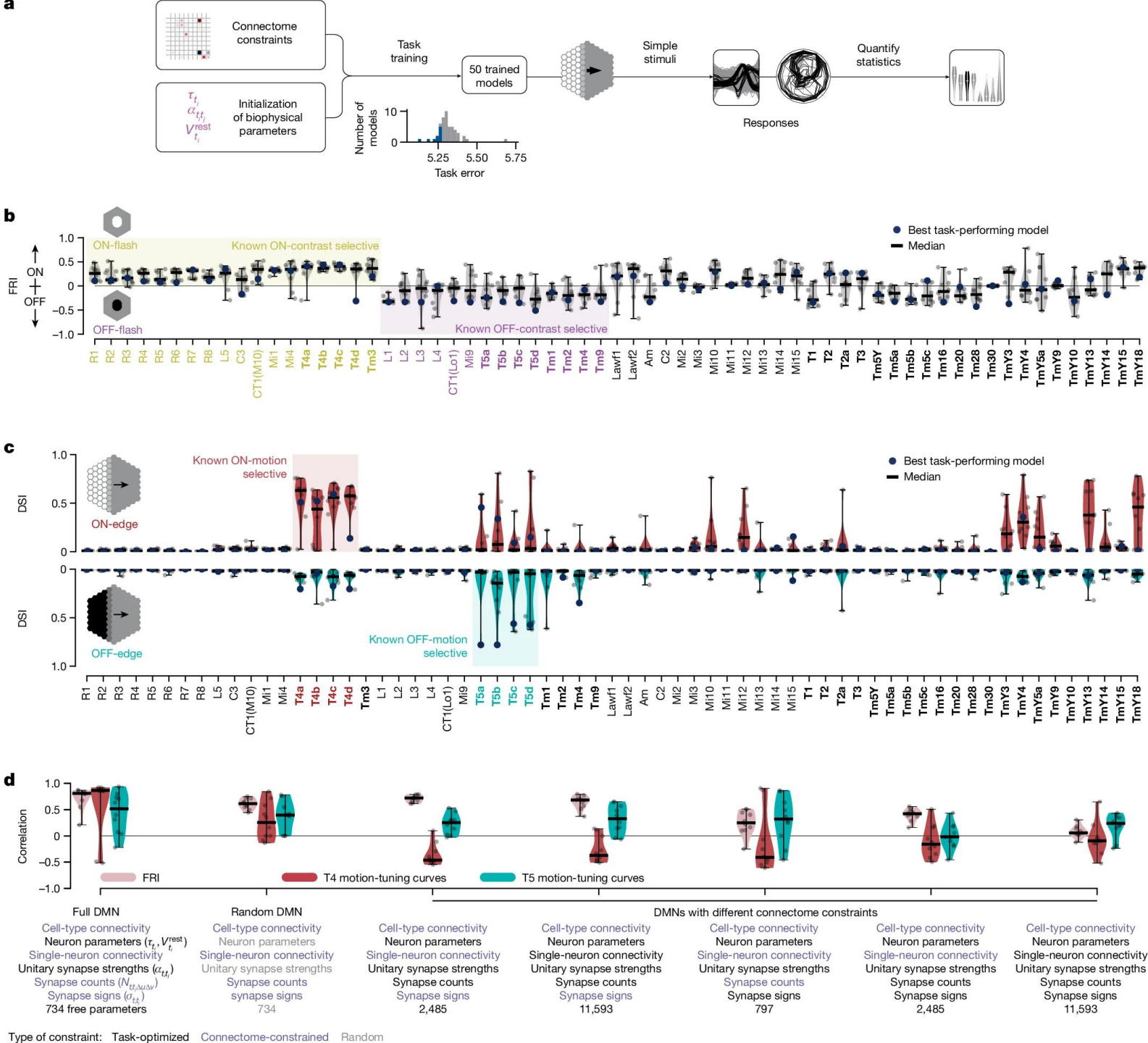
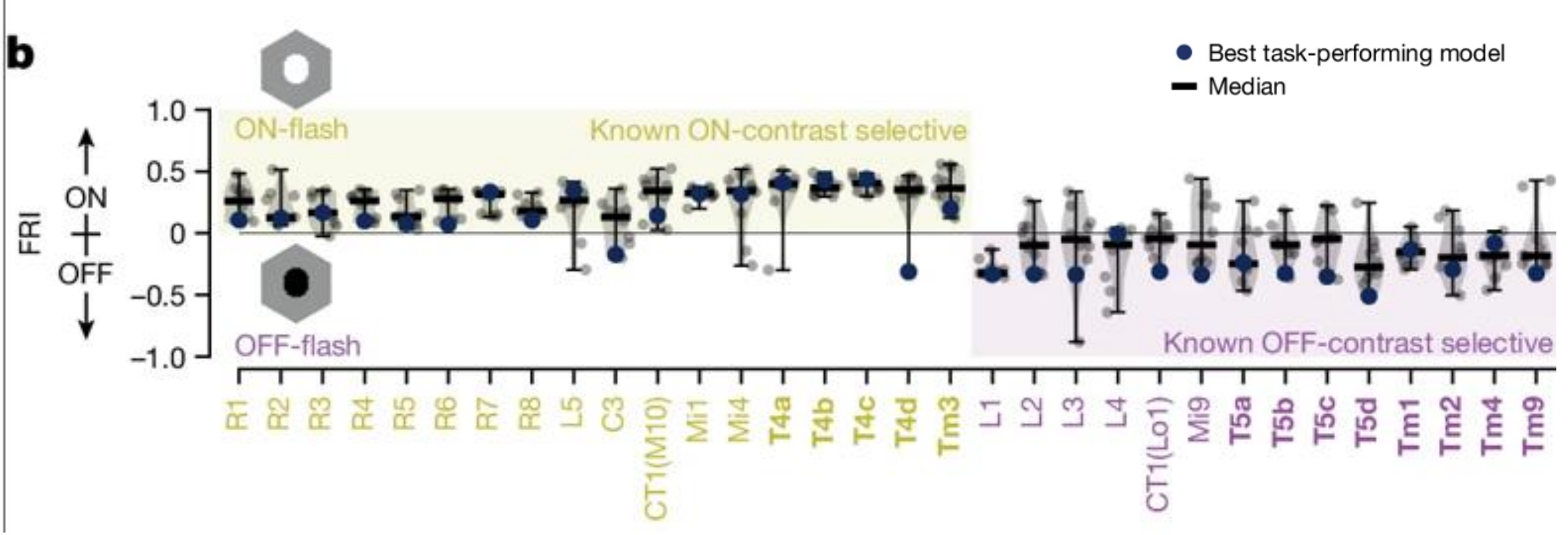


“Connectome-constrained networks predict neural activity across the fly visual system”

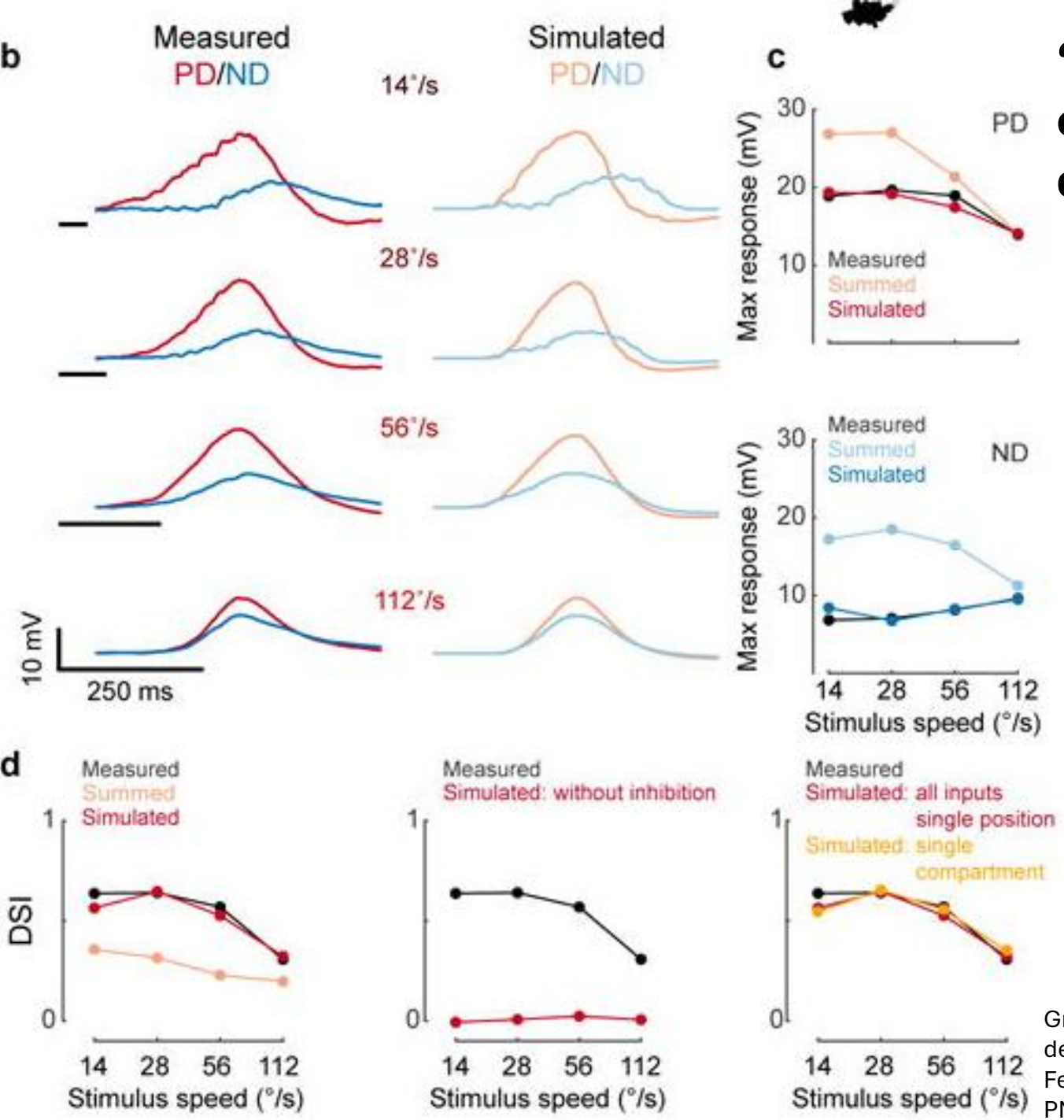
- Built many models >50
- ALL models simplified the neurons to a point
- Compared to >20 real neural recording studies
- Their “best 10” models were taken as accurate



Lappalainen, J.K., Tschopp, F.D., Prakhya, S. *et al*. Connectome-constrained networks predict neural activity across the fly visual system. *Nature* **634**, 1132–1140 (2024). https://doi.org/10.1038/s41586-024-07939-3



I don’t know brains, but I KNOW THAT THE DOTS ARE ALMOST ALL WITHIN THE ERROR BARS SO YAY VERY GOOD FOR THEM!



"Simple integration of fast excitation and offset, delayed inhibition computes directional selectivity in Drosophila."

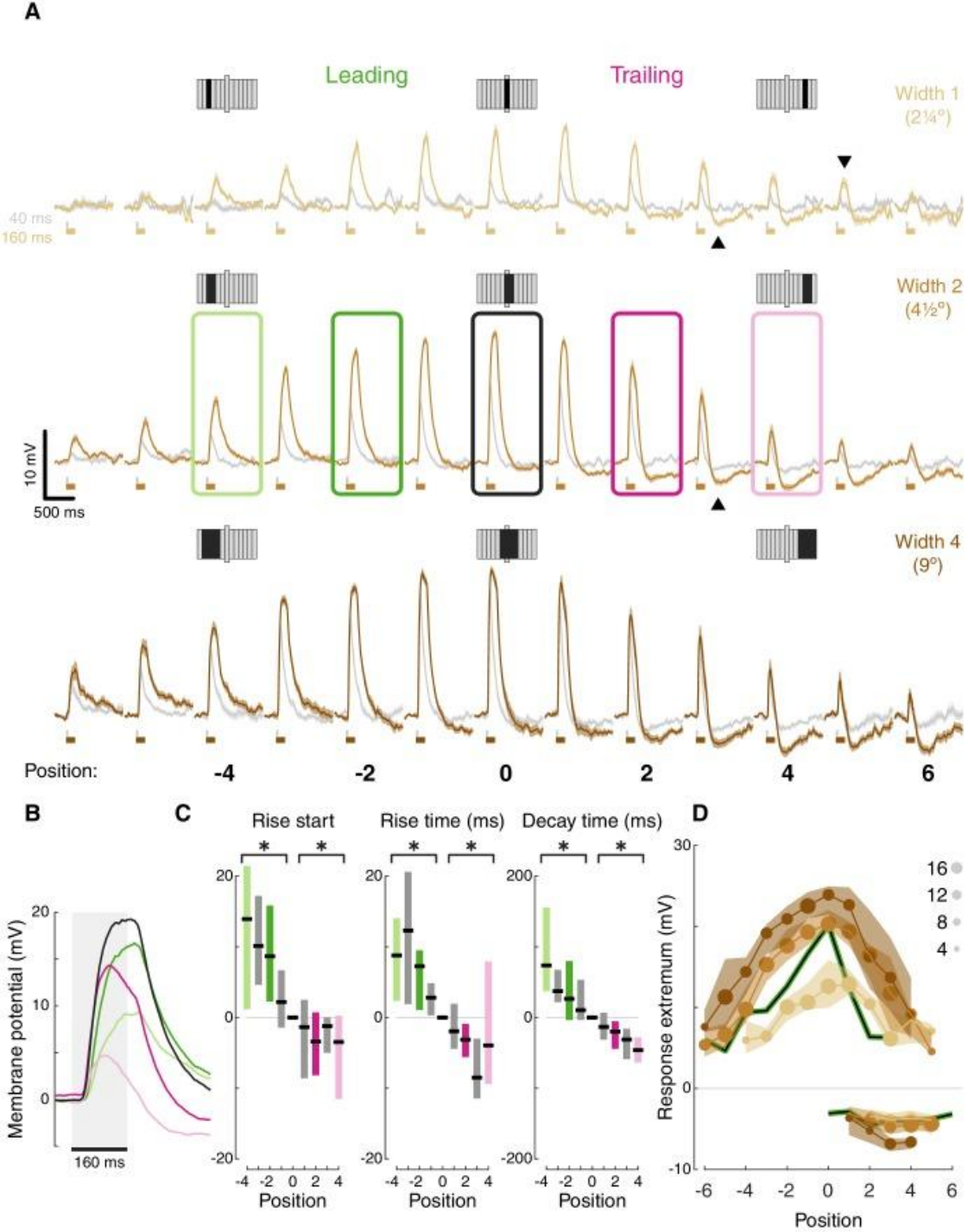
Modelled a Neuron's behaviour as:

1. Excitation
2. Delayed Inhibition

It matches very closely with measured visual motion stimuli data.

Motion wrong way: delayed inhibition cancels the signal. (blocks immediately)
Motion right way: delayed inhibition doesn't reach the signal. (too slow)

Gruntman E, Romani S, Reiser MB. Simple integration of fast excitation and offset, delayed inhibition computes directional selectivity in Drosophila. Nat Neurosci. 2018 Feb;21(2):250-257. doi: 10.1038/s41593-017-0046-4. Epub 2018 Jan 8. PMID: 29311742; PMCID: PMC5967973.



“ The computation of directional selectivity in the *Drosophila* OFF motion pathway.”

Same as last paper but for OFF (T5)

AKA motion detection of things getting dimmer.

Gruntman E, Romani S, Reiser MB. The computation of directional selectivity in the *Drosophila* OFF motion pathway. Elife. 2019 Dec 11;8:e50706. doi: 10.7554/eLife.50706. PMID: 31825313; PMCID: PMC6917495.

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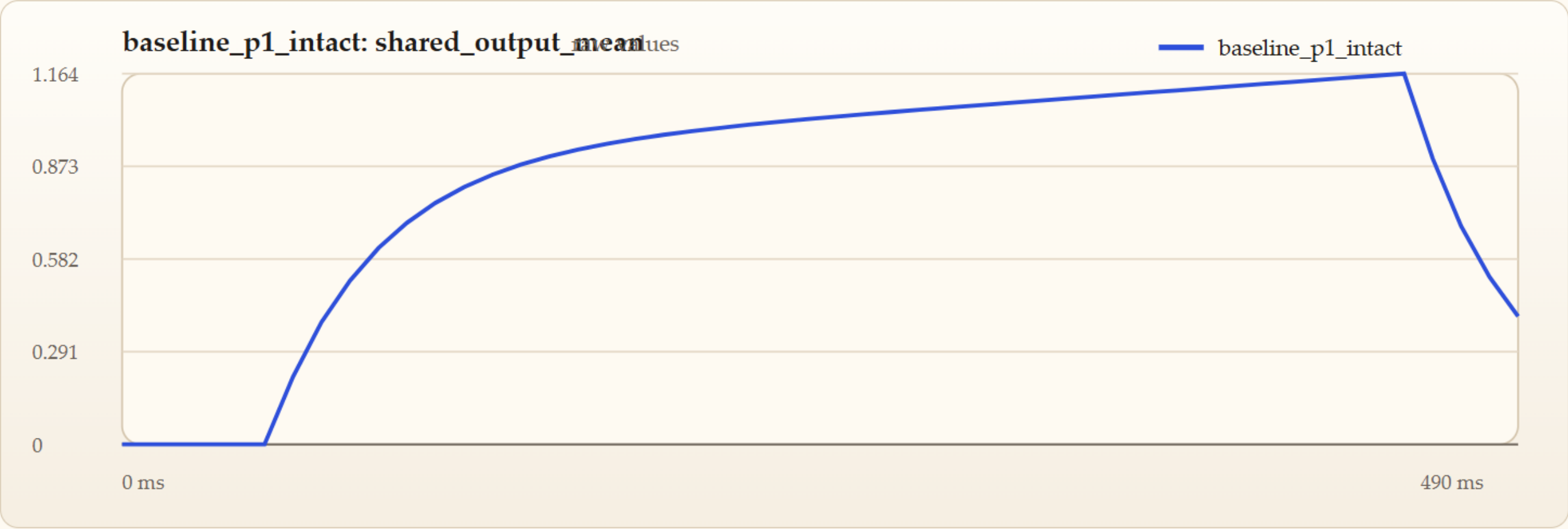
What have we achieved?

- Built a working research framework in under 3 days
- Defined a fruit fly visual circuit for the first real simulation
- Built the data, geometry, and operator pipeline/framework
- Implemented both a baseline model and a surface-wave model
- Added tests, readiness checks, and comparison tools
- 63,776 lines of python (~ 2,000 in scripts) (Thank you ai and previously written codes of brain)
  - ChatGPT, Codex, Copilot, Flywire

Why do it matter?

- This is not **just** an idea anymore
- It is now a reproducible, testable experiment
- We can directly compare standard neural modeling vs surface-wave dynamics

Results! Or something, maybe, idk....



We have a working baseline of a 2-neuron system

Results! Or something, maybe, idk....

Metric Comparison

METRIC	READOUT	WINDOW	UNITS	BASELINE_P1_INTACT	SURFACE_WAVE_INTACT
`final_endpoint_value`	`shared_output_mean`	finalized_endpoint	activation_au	0.308	2.076e+76
`sample_max_value`	`shared_output_mean`	declared_timebase	activation_au	1.164	4.152e+74
`sample_mean_value`	`shared_output_mean`	declared_timebase	activation_au	0.8198	8.473e+72
`sample_min_value`	`shared_output_mean`	declared_timebase	activation_au	0	0
`sample_peak_time_ms`	`shared_output_mean`	declared_timebase	ms	450	490

This issue is likely due to only having coded 2 neurons – should we finish more, then results will be updated. Currently 8:37 am.